

Signal Validation: gld_player_signals_daily

- **What:** every column of the signals table that the features read, checked against the daily tables, for brands 14, 23, 64, 72, 73.
- **Data:** month-start snapshots (brand 14: 2025-07 to 2026-08; brand 23: 2025-09 to 2026-09; brand 64: 2025-07 to 2026-10; brand 72: 2025-09 to 2026-09; brand 73: 2025-07 to 2026-08); on each, the players with a bet in the 30 days before. Local files only.

1. Result

19 of 28 columns pass every check in every brand. 9 fail; the models read active_days_130d, active_days_190d.

Cells: the months where the check fails and how (mismatch: the table differs from the recomputation; leak: it equals the value of 7 days later; empty: 0 for every player), and the median share of players that pass in those months.

Column	Check that fails	Brand 14	Brand 23	Brand 64	Brand 72	Brand 73	Why	Impact on the models
active_days_130d	recomputed from the daily tables	OK	OK	OK	mismatch 2026-07 to 2026-08 (89% pass)	mismatch 2026-06 (93% pass)	Higher than the bet days, never lower: it counts some days with no bet (not the days with a payment). Definition to confirm.	active_days_130d is in the model of brands 23, 73
active_days_17d	recomputed from the daily tables	OK	OK	OK	mismatch 2026-07 (92% pass)	OK	Higher than the bet days, never lower: it counts some days with no bet (not the days with a payment). Definition to confirm.	active_days_17d is in no current model
active_days_190d	recomputed from the daily tables	OK	mismatch 2025-10 (95% pass)	mismatch 2026-06; mismatch 2026-08 to 2026-09 (95% pass)	mismatch 2026-01; mismatch 2026-05; mismatch 2026-07 to 2026-09 (93% pass)	mismatch 2025-07 to 2025-10; mismatch 2026-06 (94% pass)	Higher than the bet days, never lower: it counts some days with no bet (not the days with a payment). Definition to confirm.	active_days_190d is in the model of brands 14, 64, 72, 73
days_since_bet	recomputed from the daily tables	mismatch 2025-07 to 2026-07; leak 2026-08 (21% pass)	mismatch 2025-09 to 2026-07; leak 2026-08 to 2026-09 (21% pass)	mismatch 2025-07 to 2026-07; leak 2026-08 to 2026-10 (17% pass)	mismatch 2025-09 to 2026-07; leak 2026-08 to 2026-09 (12% pass)	mismatch 2025-07 to 2026-07; leak 2026-08 (18% pass)	0 for almost every player in the backfilled months (never filled); from August 2026 it is the recency 7 days after its date (built with later data).	None: replaced by days_since_last_bet
deposit_frequency_score	filled (not 0 for every player)	empty 2025-07 to 2025-12	empty 2025-09 to 2025-12	empty 2025-07 to 2025-12	empty 2025-09 to 2025-12	empty 2025-07 to 2025-12	The payments table has almost no deposits before 2026.	deposit_frequency_score is in no current model
prior_active_days_130d	recomputed from the daily tables	OK	OK	OK	mismatch 2026-08 (92% pass)	OK	Higher than the bet days, never lower: it counts some days with no bet (not the days with a payment). Definition to confirm.	None: no feature reads it
prior_active_days_17d	recomputed from the daily tables	OK	OK	OK	mismatch 2026-07 (90% pass)	OK	Higher than the bet days, never lower: it counts some days with no bet (not the days with a payment). Definition to confirm.	prior_active_days_17d is in no current model

Column	Check that fails	Brand 14	Brand 23	Brand 64	Brand 72	Brand 73	Why	Impact on the models
tenure_days	>= days since the first bet seen in the activity	fails 2025-07 to 2026-07 (25% pass)	fails 2025-09 to 2026-07 (32% pass)	fails 2025-07 to 2026-07 (26% pass)	fails 2025-09 to 2026-07 (19% pass)	fails 2025-07 to 2026-07 (31% pass)	Shorter than the activity history in the backfilled months.	None: replaced by <code>days_since_first_bet</code>
tenure_days	grows by the days between two snapshots	fails 2025-08 to 2026-08 (25% pass)	fails 2025-10 to 2026-08 (24% pass)	fails 2025-08 to 2026-08 (19% pass)	fails 2025-10 to 2026-08 (16% pass)	fails 2025-08 to 2026-08 (23% pass)	Does not grow day by day in the backfilled months; jumps on 2026-08-01, correct after that.	None: replaced by <code>days_since_first_bet</code>
tp_night_index	between 0 and 1 (a share of play)	fails 2026-08 (85% pass)	fails 2026-08 to 2026-09 (91% pass)	fails 2026-08 to 2026-10 (88% pass)	fails 2026-08 to 2026-09 (86% pass)	fails 2026-08 (89% pass)	Reaches values above 1 once filled: not a share of play. Definition to confirm.	night_play_index is in no current model
tp_night_index	filled (not 0 for every player)	empty 2025-07 to 2026-07	empty 2025-09 to 2026-07	empty 2025-07 to 2026-07	empty 2025-09 to 2026-07	empty 2025-07 to 2026-07	Not computed before August 2026 (not backfilled).	night_play_index is in no current model

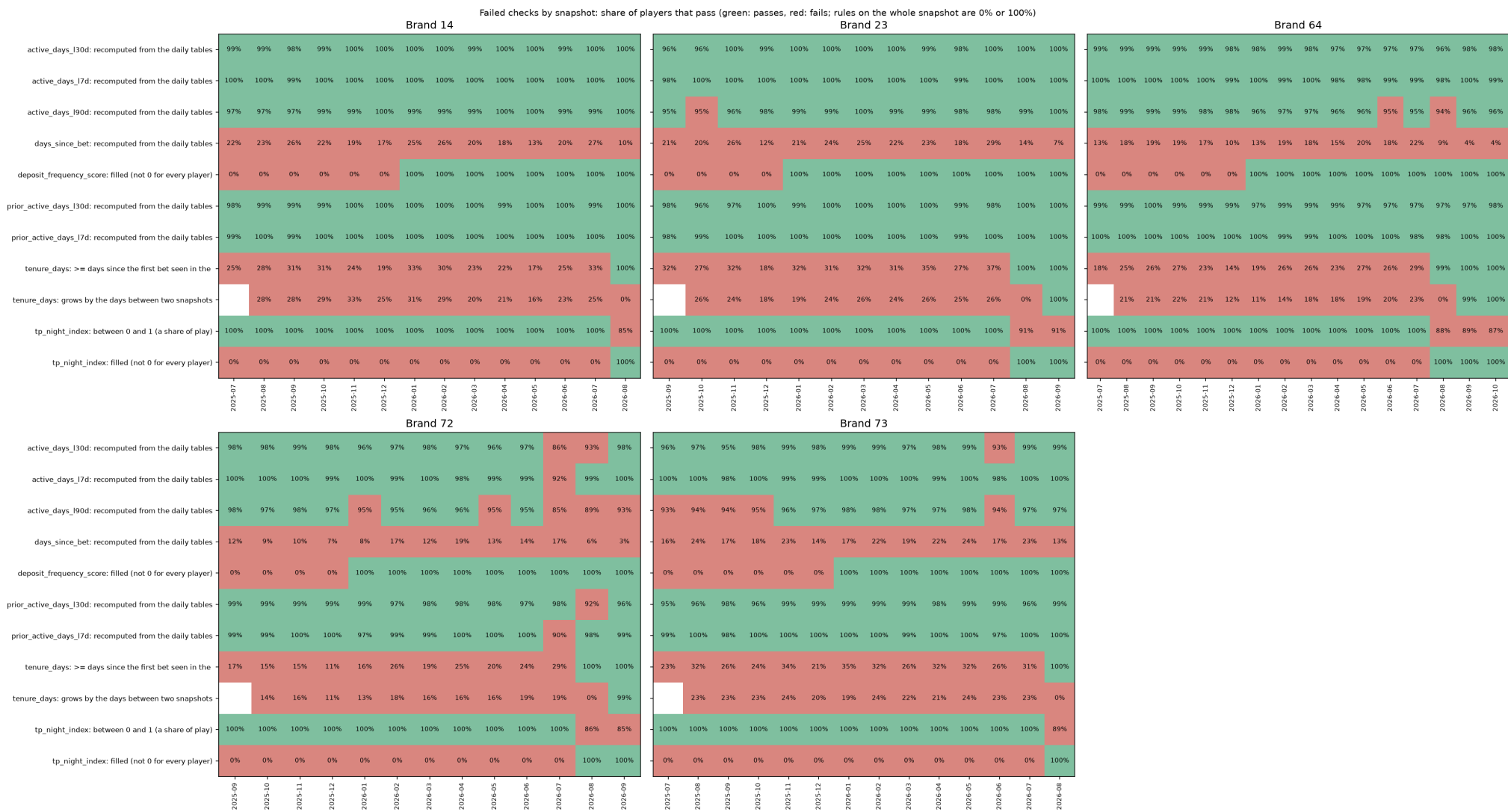


Figure 1: Failed checks by brand and snapshot

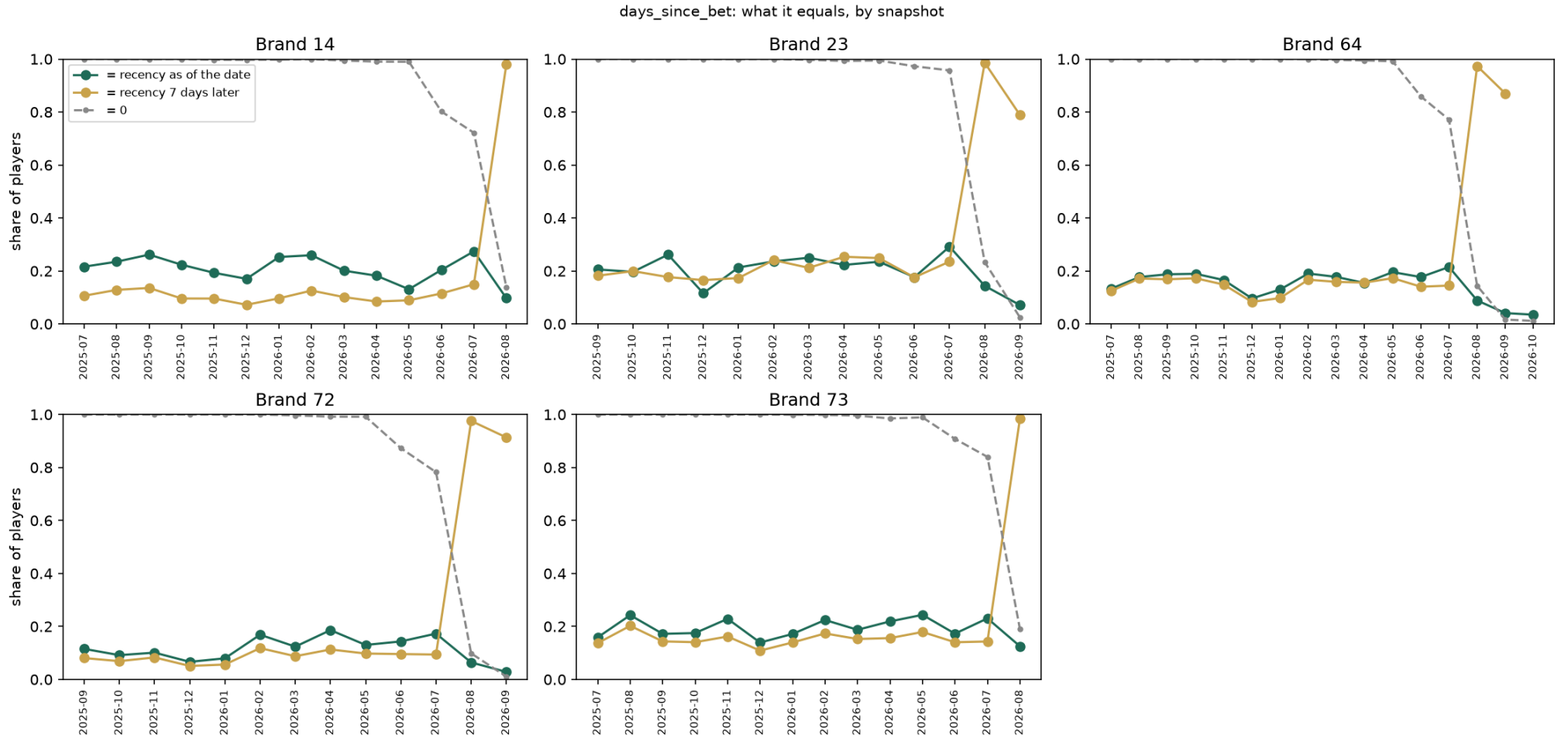


Figure 2: days_since_bet by brand

Pass in every brand: avg_bet_eur_130d, bets_130d, bets_17d, deposits_eur_130d, engagement_score, games_breadth_130d, ggr_eur_130d, ggr_eur_17d, ggr_eur_190d, prior_bets_17d, prior_ggr_eur_130d, prior_wagered_eur_17d, sessions_130d, sessions_17d, wagered_bonus_eur_130d, wagered_eur_130d, wagered_eur_17d, wagered_eur_190d, withdrawals_eur_130d.

2. Method

Check	Columns	How	Passes when
Recomputation	21 sums and counts (active days, bets, wagered, GGR, deposits, recency)	Rebuild each player's value from the daily tables (activity, financial, payments) with the window of its name	95% of the players within 2% (or 0.5) on every snapshot
Later data (leak)	The same	The same recomputation as of 7 days after the date	The table does not match the later value better (by 5 points)
Plausibility	6 that cannot be rebuilt (sessions, games, night play, scores)	Valid ranges, sessions $\geq$ active days, engagement_score = its documented formula	95% of the players on every snapshot
Filled	The same	Not 0 for every player	Every snapshot
Later data, 7-day columns	sessions_17d , deposit_frequency_score	Rank correlation with “bet (deposit) on day d”, day by day: it drops where the window ends	The window ends at most 2 days after the date, on every snapshot
Later data, 30-day columns	Sessions, engagement, games, night play (30 days)	Rank correlation with the bet days of its own 30 days and of the 30 days ending 7 days later	Its own 30 days win, on every snapshot
Tenure	tenure_days	Grows by the days between two snapshots; $\geq$ the days since the first bet in the activity	95% of the players on every snapshot

3. Actions

- **Pipeline:** it never reads **days_since_bet** or **tenure_days**; it recomputes recency (**days_since_last_bet**) and tenure (**days_since_first_bet**) from the daily activity. Tenure is correct since September 2026, but the models train on the earlier months.

4. Limitations

- It has only been tested for five brands.

Appendix: How Each Column Was Recomputed

A. Definitions. Snapshot of day D: data up to and including D, for the players with a day with bets > 0 in (D-30, D]. A value matches when it is within 2% (or 0.5) of the recomputed one. Cached daily tables, one row per player and day, read from S3 ([org/40-gold](#)).

Columns	Gold table	Value	Window
active_days_17d, active_days_130d, active_days_190d, prior_active_days_17d, prior_active_days_130d	gld_player_gaming_daily	distinct days with bets > 0	(D-7, D], (D-30, D], (D-90, D], (D-14, D-7], (D-60, D-30]
bets_17d, bets_130d, prior_bets_17d	gld_player_gaming_daily	sum of bets	(D-7, D], (D-30, D], (D-14, D-7]
wagered_eur_17d, wagered_eur_130d, wagered_eur_190d, prior_wagered_eur_17d	gld_player_gaming_daily	sum of turnover_eur	(D-7, D], (D-30, D], (D-90, D], (D-14, D-7]
ggr_eur_17d, ggr_eur_130d, ggr_eur_190d, prior_ggr_eur_130d	gld_player_gaming_daily	sum of ggr_eur	(D-7, D], (D-30, D], (D-90, D], (D-60, D-30]
wagered_bonus_eur_130d	gld_player_financial_daily	sum of wagered_bonus_eur_today	(D-30, D]
deposits_eur_130d	gld_player_payments_daily	sum of deposit_amount_eur	(D-30, D]
withdrawals_eur_130d	gld_player_payments_daily	sum of withdraw_amount_eur	(D-30, D]
avg_bet_eur_130d	gld_player_gaming_daily	sum of turnover_eur / sum of bets	(D-30, D]
days_since_bet	gld_player_gaming_daily	D minus the last day with bets > 0, up to D	all history up to D

B. Examples. One player per failing column, in the brand and snapshot where it fails most.

Column	Brand	Snapshot	tenant_id	player_id	Table	Recomputed	Recomputed 7 days later
active_days_17d	72	2026-07-01	9a13a78d-ec27-4f7c-8f75-b89575438980	3267253	6	3	
active_days_130d	72	2026-07-01	9a13a78d-ec27-4f7c-8f75-b89575438980	3267253	17	7	
active_days_190d	72	2026-07-01	9a13a78d-ec27-4f7c-8f75-b89575438980	3267253	18	8	
prior_active_days_17d	72	2026-07-01	9a13a78d-ec27-4f7c-8f75-b89575438980	3267253	7	4	
prior_active_days_130d	72	2026-08-01	9a13a78d-ec27-4f7c-8f75-b89575438980	3267253	17	7	
days_since_bet	72	2025-12-01	9a13a78d-ec27-4f7c-8f75-b89575438980	3220036	0	23	30
days_since_bet	72	2026-09-01	9a13a78d-ec27-4f7c-8f75-b89575438980	3263431	16	9	16
tenure_days (vs days since the first bet)	72	2025-12-01	9a13a78d-ec27-4f7c-8f75-b89575438980	3220036	193	216	
tp_night_index (above 1)	14	2026-08-01	73c25416-d9de-46fb-8a73-2e64f63d6330	109490	1.29		